A Talk on Double Affine Hecke Algebras

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July 16, 2026

The purpose of this note is to introduce the X–Y duality of double affine Hecke algebras and to illustrate it explicitly in the duality $B_n^{(1)} \leftrightarrow C_n^{(1)}$.

Main Statements

Statement (Root-system and DAHA involution). *For Ion's dual affine root system R^ι,*

$$\alpha_j^\iota = \frac{A_j}{\sqrt{p}}, \qquad \mathring{Q}^\iota = \frac{1}{\sqrt{p}} M, \qquad M^\iota = \sqrt{p}\, \mathring{Q}.$$

There is a semilinear algebra isomorphism

$$\phi_{\widetilde{W}} : \mathcal{H}_{\widetilde{W}} \longrightarrow \mathcal{H}_{\widetilde{W}^\iota}$$

characterized by

$$\phi_{\widetilde{W}}(T_j) = (T_j^\iota)^{-1}, \qquad \phi_{\widetilde{W}}(Y_\mu) = X^\iota_{\mu/\sqrt{p}}, \qquad \phi_{\widetilde{W}}(X_\beta) = Y^\iota_{\sqrt{p}\,\beta},$$

together with

$$\phi_{\widetilde{W}}(q) = (q^\iota)^{-1}, \qquad \phi_{\widetilde{W}}(t_j^{1/2}) = (t_j^\iota)^{-1/2}.$$

In particular, $(B_n^{(1)})^\iota = C_n^{(1)}$ and $p = 2$.

Statement (Non-symmetric Macdonald polynomials). *For generic parameters and every $\lambda \in \mathring{Q}$, there is a unique monic triangular common Y-eigenvector*

$$E_\lambda^R = X_\lambda + \sum_{\eta < \lambda} c^R_{\lambda\eta} X_\eta, \qquad Y_\mu E_\lambda^R = \gamma_\lambda^R(\mu) E_\lambda^R \quad (\mu \in M).$$

Statement (Non-symmetric evaluation-duality). *Let*

$$X^\iota_{\mu/\sqrt{p}}\big(\mathrm{sp}_R(\lambda)\big) = \gamma_\lambda^R(-\mu), \qquad X_\lambda\big(\mathrm{sp}_{R^\iota}(\beta)\big) = \gamma_\beta^{R^\iota}(-\sqrt{p}\,\lambda),$$

and normalize E_λ^R and $E_\beta^{R^\iota}$ at the corresponding vacuum spectral points. Then

$$\boxed{\mathcal{E}_\lambda^R\big(\mathrm{sp}_{R^\iota}(\beta)\big) = \mathcal{E}_\beta^{R^\iota}\big(\mathrm{sp}_R(\lambda)\big).}$$

All conventions, proofs, recursive constructions, and the explicit $B_n^{(1)}$–$C_n^{(1)}$ computation are given in the following pages.

Contents

1 Notation and Conventions

We follow Ion's conventions throughout; in particular, $X_\delta = q^{-1}$, the X-index lattice is the finite root lattice $\mathring{Q}$, and the Y-index lattice is Ion's translation lattice M. The rightmost factor in a product of operators acts first.

Symbol	Meaning
$R,\ R^\iota$	An affine root system and its Ion dual.
$\alpha_0, \ldots, \alpha_n$	The affine simple roots; $\alpha_1, \ldots, \alpha_n$ form the finite system.
$\delta,\ \theta$	The null root and the highest finite root, with $\alpha_0 = \delta - \theta$.
$\mathring{Q}$	The finite root lattice, used to index the X-monomials.
M	Ion's translation lattice, generated by $A_j = e_j \alpha_j$, used to index the Y-operators.
$\mathring{W},\ W$	The finite and affine Weyl groups.
λ_μ	The translation indexed by $\mu \in M$; at level one it acts by $x \mapsto x + \mu$.
$X_\beta,\ Y_\mu$	The two commuting lattice directions of DAHA, indexed by $\beta \in \mathring{Q}$ and $\mu \in M$.
T_j	A Hecke generator corresponding to the simple reflection s_j.
$p = e_s$	Ion's lacing number; $p = 2$ for $B_n^{(1)}$ and $C_n^{(1)}$.
$\phi_{\widetilde{W}}$	Ion's semilinear DAHA isomorphism from R to R^ι.
ω_R	The parameter-preserving Fourier map $\phi_{\widetilde{W}} \circ \overline{(\cdot)}$.
$\gamma_\lambda^R(\mu)$	The Y_μ-eigenvalue of the non-symmetric Macdonald polynomial E_λ^R.
$\mathrm{sp}_R(\lambda)$	The spectral character on the dual polynomial algebra associated with λ.
$E_\lambda^R,\ \mathcal{E}_\lambda^R$	The Ion-normalized non-symmetric Macdonald polynomial and its vacuum-normalized version.

When $R = B_n^{(1)}$, the notation E_λ^R is the same as the notation E_λ^{I} used when emphasizing Ion normalization.

2 Root Data

Let

$$A = (a_{ij})_{0 \le i,j \le n}$$

be an irreducible affine Cartan matrix, and let

$$\mathring{A} = (a_{ij})_{1 \le i,j \le n}$$

be its finite part. Let

$$\alpha_0, \alpha_1, \ldots, \alpha_n$$

be the simple affine roots, and let

$$\mathring{Q} = \bigoplus_{j=1}^{n} \mathbb{Z}\alpha_j$$

be the finite root lattice. Write

$$\delta = \sum_{j=0}^{n} a_j \alpha_j, \qquad \delta^\vee = \sum_{j=0}^{n} a_j^\vee \alpha_j^\vee. \tag{1}$$

We exclude type $A_{2n}^{(2)}$. Thus $a_0 = 1$, and

$$\alpha_0 = \delta - \theta, \tag{2}$$

where θ is the highest root of the finite root system.

For $1 \le j \le n$, set

$$d_j := \frac{a_j}{a_j^\vee}, \qquad e_j := \max\{d_j, a_0^{-1}\}, \qquad A_j := e_j \alpha_j, \tag{3}$$

and define

$$M := \bigoplus_{j=1}^{n} \mathbb{Z} A_j. \tag{4}$$

We regard M as an abelian group of translations $\{\lambda_\mu \mid \mu \in M\}$. The number e_j depends only on the length of α_j. We write e_s and e_l for its values on short and long roots, respectively, and set

$$p := e_s. \tag{5}$$

In the simply laced case, $p = 1$.

We use the symmetric bilinear form

$$(\alpha_j, \alpha_k) := d_j^{-1} a_{jk}, \qquad 0 \le j, k \le n, \tag{6}$$

and the reflections

$$s_\alpha(x) = x - \frac{2(x, \alpha)}{(\alpha, \alpha)} \alpha. \tag{7}$$

The finite Weyl group has the Coxeter presentation

$$\mathring{W} = \left\langle s_i \; (1 \le i \le n) \;\middle|\; s_i^2 = 1,\ \underbrace{s_i s_j s_i \cdots}_{m_{ij} \text{ factors}} = \underbrace{s_j s_i s_j \cdots}_{m_{ij} \text{ factors}} \; (i \ne j) \right\rangle, \tag{8}$$

where

$$m_{ij} = \begin{cases} 2, & a_{ij} a_{ji} = 0, \\ 3, & a_{ij} a_{ji} = 1, \\ 4, & a_{ij} a_{ji} = 2, \\ 6, & a_{ij} a_{ji} = 3, \\ \infty, & a_{ij} a_{ji} \ge 4. \end{cases} \tag{9}$$

See [Kac83, §3.13]. The affine Weyl group is

$$W = \mathring{W} \ltimes M. \tag{10}$$

3 The Double Affine Hecke Algebra

Let m be the least common denominator of

$$\{(\alpha_j, \Lambda_k) \mid 1 \le j, k \le n\},$$

where Λ_k is the k-th fundamental weight, and let $\mathbb{F}$ be the field of rational functions in $q^{1/m}$ and the $t_j^{1/2}$.

The two lattice algebras are

$$\mathbb{F}[M] = \bigoplus_{\mu \in M} \mathbb{F} Y_\mu, \qquad \mathbb{F}[\mathring{Q}] = \bigoplus_{\beta \in \mathring{Q}} \mathbb{F} X_\beta, \tag{11}$$

with

$$Y_\mu Y_\nu = Y_{\mu+\nu}, \qquad X_\beta X_\gamma = X_{\beta+\gamma}. \tag{12}$$

The parameters are constant on root-length orbits:

$$d_j = d_k \quad \Longleftrightarrow \quad t_j = t_k. \tag{13}$$

3.1 The finite Hecke algebra

The finite Hecke algebra $\mathcal{H}_{\mathring{W}}$ is generated by $T_1, \ldots, T_n$, subject to the braid relations corresponding to (8) and the quadratic relations

$$T_j - T_j^{-1} = t_j^{1/2} - t_j^{-1/2}. \tag{14}$$

3.2 The affine Hecke algebra

The affine Hecke algebra $\mathcal{H}_W$ is generated by $\mathcal{H}_{\mathring{W}}$ and $\mathbb{F}[M]$, subject to

$$Y_\mu T_j - T_j Y_{s_j(\mu)} = \left(t_j^{1/2} - t_j^{-1/2}\right) \frac{Y_\mu - Y_{s_j(\mu)}}{1 - Y_{A_j}}, \tag{15}$$

for $\mu \in M$ and $1 \le j \le n$.

3.3 The double affine Hecke algebra

Definition 3.1. *The double affine Hecke algebra $\mathcal{H}_{\widetilde{W}}$ is generated by $\mathcal{H}_W$ and $\mathbb{F}[\mathring{Q}]$, subject to*

$$T_j X_\beta - X_{s_j(\beta)} T_j = \left(t_j^{1/2} - t_j^{-1/2}\right) \frac{X_\beta - X_{s_j(\beta)}}{1 - X_{-\alpha_j}}, \tag{16}$$

for $\beta \in \mathring{Q}$ and $0 \le j \le n$, together with

$$X_\delta = q^{-1}. \tag{17}$$

For $j = 0$, the monomial $X_{s_0(\beta)}$ is interpreted using $X_\delta = q^{-1}$.

Thus $\mathcal{H}_{\widetilde{W}}$ contains two commuting lattice algebras:

$$\mathbb{F}[\mathring{Q}] \quad \text{generated by the } X_\beta,$$

and

$$\mathbb{F}[M] \quad \text{generated by the } Y_\mu.$$

The involution below exchanges these two directions.

4 The Root-System and DAHA Involutions

4.1 The root-system involution

Definition 4.1. *Define a new affine root system R^ι by*

$$\alpha_j^\iota := \frac{A_j}{\sqrt{p}} = \frac{e_j}{\sqrt{p}}\alpha_j, \qquad 1 \le j \le n, \tag{18}$$

together with

$$\delta^\iota = \delta. \tag{19}$$

The affine simple root α_0^ι is determined by the affine type.

Under the natural identification of the finite root spaces,

$$\mathring{Q}^\iota = \frac{1}{\sqrt{p}}M, \qquad M^\iota = \sqrt{p}\,\mathring{Q}. \tag{20}$$

The involution preserves the affine type in the simply laced cases. In particular,

$$A_n^{(1)\,\iota} = A_n^{(1)}. \tag{21}$$

In the non-simply-laced classical cases,

$$B_n^{(1)\,\iota} = C_n^{(1)}, \qquad C_n^{(1)\,\iota} = B_n^{(1)}. \tag{22}$$

On finite roots, ι interchanges short and long roots.

4.2 Ion's semilinear DAHA isomorphism

Let $\mathcal{H}_{\widetilde{W}^\iota}$ be the DAHA associated with R^ι. Its generators are denoted by

$$T_j^\iota, \qquad X_\gamma^\iota, \qquad Y_\nu^\iota.$$

Theorem 4.2 (Ion's DAHA involution). *The assignments*

$$\phi_{\widetilde{W}}(T_j) = (T_j^\iota)^{-1}, \qquad 1 \le j \le n, \tag{23}$$

$$\phi_{\widetilde{W}}(Y_\mu) = X^\iota_{\mu/\sqrt{p}}, \qquad \mu \in M, \tag{24}$$

$$\phi_{\widetilde{W}}(X_\beta) = Y^\iota_{\sqrt{p}\,\beta}, \qquad \beta \in \mathring{Q}, \tag{25}$$

extend to a semilinear algebra isomorphism

$$\phi_{\widetilde{W}} : \mathcal{H}_{\widetilde{W}} \longrightarrow \mathcal{H}_{\widetilde{W}^\iota}, \tag{26}$$

with

$$\phi_{\widetilde{W}}(q) = (q^\iota)^{-1}, \qquad \phi_{\widetilde{W}}(t_j^{1/2}) = (t_j^\iota)^{-1/2}. \tag{27}$$

Its inverse is $\phi_{\widetilde{W}^\iota}$.

This is Ion's Theorems 2.2–2.3 [Ion03, Theorems 2.2–2.3]. Its content is summarized by

$$\boxed{X_\beta \longleftrightarrow Y^\iota_{\sqrt{p}\,\beta}}. \tag{28}$$

Thus the DAHA involution exchanges the X-lattice and the Y-lattice, while interchanging the root system with its ι-dual.

4.3 The parameter-preserving Fourier map

Let $\overline{(\cdot)}$ denote the standard semilinear bar involution, characterized on the relevant generators by

$$\overline{q} = q^{-1}, \qquad \overline{t_j^{1/2}} = t_j^{-1/2}, \qquad \overline{T_j} = T_j^{-1}, \qquad \overline{X_\beta} = X_{-\beta}, \qquad \overline{Y_\mu} = Y_{-\mu}. \tag{29}$$

Define

$$\omega_R := \phi_{\widetilde{W}} \circ \overline{(\cdot)}. \tag{30}$$

After identifying the transported parameters, ω_R is parameter-preserving and satisfies

$$\omega_R(T_j) = T_j^\iota, \qquad \omega_R(Y_\mu) = X^\iota_{-\mu/\sqrt{p}}, \qquad \omega_R(X_\beta) = Y^\iota_{-\sqrt{p}\,\beta}. \tag{31}$$

Remark 4.3. *The sign distinction is important. Ion's semilinear map* $\phi_{\widetilde{W}}$ *sends* X_β *to* $Y^\iota_{+\sqrt{p}\beta}$*, whereas the parameter-preserving Fourier map* ω_R *used in evaluation-duality sends it to* $Y^\iota_{-\sqrt{p}\beta}$*.*

5 Non-symmetric Macdonald Polynomials

The basic polynomial representation is

$$\mathcal{P}_R := \mathbb{F}[\mathring{Q}]. \tag{32}$$

For $\lambda \in \mathring{Q}$, let λ_+ be the dominant element of $\mathring{W}\lambda$, and let $v(\lambda)$ be the shortest element of $\mathring{W}$ sending λ to the anti-dominant chamber. Define

$$\lambda \geq \eta \iff \begin{cases} \lambda_+ > \eta_+, & \text{in strict dominance order,} \\ \lambda_+ = \eta_+ \text{ and } v(\lambda) \leq v(\eta), & \text{in finite Bruhat order.} \end{cases} \tag{33}$$

Set

$$\mathring{Q}_{\leq\lambda} := \{\eta \in \mathring{Q} \mid \eta \leq \lambda\}, \qquad \mathcal{P}_{\leq\lambda} := \mathrm{span}_{\mathbb{F}}\{X_\eta \mid \eta \leq \lambda\}. \tag{34}$$

The set $\mathring{Q}_{\leq\lambda}$ is finite and lower closed.

Theorem 5.1 (Y-triangularity). *For every $\mu \in M$ and $\lambda \in \mathring{Q}$, there are unique coefficients $\gamma^R_\lambda(\mu) \in \mathbb{F}^\times$ and $a^R_{\eta\lambda}(\mu) \in \mathbb{F}$ such that*

$$Y_\mu X_\lambda = \gamma^R_\lambda(\mu) X_\lambda + \sum_{\eta < \lambda} a^R_{\eta\lambda}(\mu) X_\eta. \tag{35}$$

In particular, $\mathcal{P}_{\le\lambda}$ is invariant under every Y_μ.

For anti-dominant μ, the theorem follows by expanding a reduced expression of the positive braid lift T_{λ_μ}. For a general $\mu = \mu_1 - \mu_2$, with μ_1, μ_2 anti-dominant, one uses $Y_\mu = Y_{\mu_1} Y_{\mu_2}^{-1}$; the inverse of an invertible lower triangular operator is again lower triangular. This is Macdonald's triangularity argument written in Ion's $\mathring{Q}, M, X, Y$ notation [Mac03, 4.6.4–4.6.12].

Definition 5.2. *For generic parameters, the Ion-normalized non-symmetric Macdonald polynomial $E^R_\lambda \in \mathcal{P}_R$ is the monic common eigenvector satisfying*

$$E^R_\lambda = X_\lambda + \sum_{\eta < \lambda} c^R_{\lambda\eta} X_\eta, \tag{36}$$

$$Y_\mu E^R_\lambda = \gamma^R_\lambda(\mu) E^R_\lambda, \qquad \mu \in M. \tag{37}$$

Here "generic" means that the joint characters γ^R_η are pairwise distinct on the finite lower ideal under consideration and that the denominators below do not vanish.

Proposition 5.3 (Finite triangular recursion). *Choose $\nu \in M$ such that $\gamma^R_\eta(\nu)$, $\eta \le \lambda$, are pairwise distinct, and write*

$$Y_\nu X_\zeta = \gamma^R_\zeta(\nu) X_\zeta + \sum_{\eta < \zeta} a^R_{\eta\zeta}(\nu) X_\eta. \tag{38}$$

Then

$$\boxed{c^R_{\lambda\eta} = \frac{a^R_{\eta\lambda}(\nu) + \displaystyle\sum_{\eta < \zeta < \lambda} c^R_{\lambda\zeta} a^R_{\eta\zeta}(\nu)}{\gamma^R_\lambda(\nu) - \gamma^R_\eta(\nu)}.} \tag{39}$$

The computation proceeds from the weights nearest to λ downward, one layer at a time.

From the polynomial representation, $T_i 1 = t_i^{1/2} 1$. Hence every Y_μ sends 1 to a scalar multiple of 1, and leading-term normalization gives

$$\boxed{E^R_0 = 1.} \tag{40}$$

6 Macdonald Evaluation-Duality

6.1 Spectral points

For $\lambda \in \mathring{Q}$, define an algebraic character on the dual polynomial algebra $\mathcal{P}_{R^\iota} = \mathbb{F}[\mathring{Q}^\iota_X]$ by

$$\mathrm{sp}_R(\lambda) : \mathcal{P}_{R^\iota} \longrightarrow \mathbb{F}, \qquad \boxed{X^\iota_{\mu/\sqrt{p}}(\mathrm{sp}_R(\lambda)) := \gamma^R_\lambda(-\mu)}, \quad \mu \in M. \tag{41}$$

This covers every index in $\mathring{Q}^\iota$, since $\mathring{Q}^\iota = p^{-1/2}M$. Similarly, for $\beta \in \mathring{Q}^\iota$, define

$$\mathrm{sp}_{R^\iota}(\beta) : \mathcal{P}_R \longrightarrow \mathbb{F}, \qquad \boxed{X_\lambda\big(\mathrm{sp}_{R^\iota}(\beta)\big) := \gamma_\beta^{R^\iota}(-\sqrt{p}\,\lambda)}, \quad \lambda \in \mathring{Q}. \tag{42}$$

Equations (41)–(42) completely define the spectral points. The minus sign follows from $\omega_R(X_\lambda) = Y^\iota_{-\sqrt{p}\lambda}$, not from an additional choice of convention.

Define the vacuum-normalized polynomials

$$\mathcal{E}_\lambda^R := \frac{E_\lambda^R}{E_\lambda^R(\mathrm{sp}_{R^\iota}(0))}, \qquad \mathcal{E}_\beta^{R^\iota} := \frac{E_\beta^{R^\iota}}{E_\beta^{R^\iota}(\mathrm{sp}_R(0))}. \tag{43}$$

We assume that both denominators are nonzero, as holds for generic parameters.

6.2 The duality theorem

For $f \in \mathcal{P}_R$ and $g \in \mathcal{P}_{R^\iota}$, define the Fourier pairing

$$[f, g]_R := \big(\omega_R(f)g\big)\big(\mathrm{sp}_R(0)\big). \tag{44}$$

The standard symmetry of this pairing is

$$[f, g]_R = [g, f]_{R^\iota}; \tag{45}$$

see [Mac03, 4.7.14].

Theorem 6.1 (General non-symmetric evaluation-duality)**.** *For every $\lambda \in \mathring{Q}$ and $\beta \in \mathring{Q}^\iota$,*

$$\boxed{\mathcal{E}_\lambda^R\big(\mathrm{sp}_{R^\iota}(\beta)\big) = \mathcal{E}_\beta^{R^\iota}\big(\mathrm{sp}_R(\lambda)\big).} \tag{46}$$

Equivalently,

$$\boxed{E_\lambda^R\big(\mathrm{sp}_{R^\iota}(\beta)\big)E_\beta^{R^\iota}\big(\mathrm{sp}_R(0)\big) = E_\lambda^R\big(\mathrm{sp}_{R^\iota}(0)\big)E_\beta^{R^\iota}\big(\mathrm{sp}_R(\lambda)\big).} \tag{47}$$

Proof. By (31) and the eigenvalue equation (37),

$$\omega_R(f)E_\beta^{R^\iota} = f\big(\mathrm{sp}_{R^\iota}(\beta)\big)E_\beta^{R^\iota}, \qquad f \in \mathcal{P}_R. \tag{48}$$

Evaluation at $\mathrm{sp}_R(0)$ gives

$$[f, E_\beta^{R^\iota}]_R = f\big(\mathrm{sp}_{R^\iota}(\beta)\big)E_\beta^{R^\iota}\big(\mathrm{sp}_R(0)\big). \tag{49}$$

Exchanging R and R^ι gives

$$[E_\lambda^R, g]_{R^\iota} = g\big(\mathrm{sp}_R(\lambda)\big)E_\lambda^R\big(\mathrm{sp}_{R^\iota}(0)\big). \tag{50}$$

Take $f = E_\lambda^R$ and $g = E_\beta^{R^\iota}$, and use (45). This proves (47); division by the two nonzero vacuum evaluations proves (46). ◻

7 Examples

7.1 Examples of the root-system involution

The finite simple roots of types A, B, and C are as follows:

$$
\begin{aligned}
A_{n-1}:&\quad \alpha_i = \varepsilon_i - \varepsilon_{i+1}, && 1 \le i < n; && \\
B_n:&\quad \alpha_i = \varepsilon_i - \varepsilon_{i+1}, && 1 \le i < n, && \alpha_n = \varepsilon_n; \\
C_n:&\quad \alpha_i = \varepsilon_i - \varepsilon_{i+1}, && 1 \le i < n, && \alpha_n = 2\varepsilon_n.
\end{aligned}
$$

Table 1: Affine labels and dual labels.

Type	$(a_0, a_1, \ldots, a_n)$	$(a_0^\vee, a_1^\vee, \ldots, a_n^\vee)$
$A_n^{(1)}$	$(1, 1, \ldots, 1)$	$(1, 1, \ldots, 1)$
$B_n^{(1)}$	$(1, 1, 2, \ldots, 2, 2)$	$(1, 1, 2, \ldots, 2, 1)$
$C_n^{(1)}$	$(1, 2, \ldots, 2, 1)$	$(1, 1, \ldots, 1)$

Consequently,

$$A_n^{(1)}: \quad M = \mathring{Q}, \tag{51}$$

$$B_n^{(1)}: \quad M = \bigoplus_{i=1}^{n-1} \mathbb{Z}\alpha_i \oplus \mathbb{Z}(2\alpha_n), \tag{52}$$

$$C_n^{(1)}: \quad M = \bigoplus_{i=1}^{n-1} \mathbb{Z}(2\alpha_i) \oplus \mathbb{Z}\alpha_n. \tag{53}$$

Now specialize to $B_n^{(1)}$. Then

$$d_1 = \cdots = d_{n-1} = 1, \qquad d_n = 2, \tag{54}$$

and hence

$$e_1 = \cdots = e_{n-1} = 1, \qquad e_n = 2. \tag{55}$$

Thus

$$A_j = \alpha_j \quad (1 \le j < n), \qquad A_n = 2\alpha_n, \qquad p = e_s = 2. \tag{56}$$

By definition,

$$\alpha_i^\iota = \frac{\varepsilon_i - \varepsilon_{i+1}}{\sqrt{2}} \quad (i < n), \qquad \alpha_n^\iota = \sqrt{2}\,\varepsilon_n. \tag{57}$$

Their squared lengths are

$$(\alpha_i^\iota, \alpha_i^\iota) = 1 \quad (i < n), \qquad (\alpha_n^\iota, \alpha_n^\iota) = 2. \tag{58}$$

Thus the first $n-1$ roots are short and the last root is long. The finite Cartan matrix is

$$\mathring{A}^\iota = \begin{pmatrix} 2 & -1 & 0 & \cdots & 0 \\ -1 & 2 & -1 & \ddots & \vdots \\ 0 & -1 & \ddots & \ddots & 0 \\ \vdots & \ddots & \ddots & 2 & -2 \\ 0 & \cdots & 0 & -1 & 2 \end{pmatrix}, \tag{59}$$

which is of type C_n. Hence

$$\boxed{(B_n^{(1)})^\iota = C_n^{(1)}.} \tag{60}$$

In this case, Theorem 4.2 becomes

$$\begin{aligned} \phi_{\widetilde{W}}(T_j) &= (T_j^\iota)^{-1}, && 1 \le j \le n, && (61)\\ \phi_{\widetilde{W}}(Y_\mu) &= X^\iota_{\mu/\sqrt{2}}, && \mu \in M, && (62)\\ \phi_{\widetilde{W}}(X_\beta) &= Y^\iota_{\sqrt{2}\beta}, && \beta \in \mathring{Q}, && (63)\\ \phi_{\widetilde{W}}(q) &= (q^\iota)^{-1}, && && (64)\\ \phi_{\widetilde{W}}(t_j) &= (t_j^\iota)^{-1}. && && (65) \end{aligned}$$

Moreover,

$$\mathring{Q}^\iota = \frac{1}{\sqrt{2}} M = \bigoplus_{i=1}^{n-1} \mathbb{Z}\frac{\varepsilon_i - \varepsilon_{i+1}}{\sqrt{2}} \oplus \mathbb{Z}(\sqrt{2}\varepsilon_n), \tag{66}$$

whereas

$$M^\iota = \bigoplus_{i=1}^{n-1} \mathbb{Z}(2\alpha_i^\iota) \oplus \mathbb{Z}\alpha_n^\iota = \sqrt{2}\,\mathring{Q}. \tag{67}$$

This shows precisely that the X-lattice of B_n becomes the Y-lattice of C_n, while the Y-lattice of B_n becomes the X-lattice of C_n.

7.2 The polynomial representation of type $B_n^{(1)}$

Let $n \ge 3$, and put

$$\alpha_i = \varepsilon_i - \varepsilon_{i+1} \quad (1 \le i < n), \qquad \alpha_n = \varepsilon_n, \tag{68}$$

and

$$\theta = \varepsilon_1 + \varepsilon_2, \qquad \alpha_0 = \delta - \theta. \tag{69}$$

Then

$$\mathring{Q} = \mathbb{Z}^n, \qquad M = \left\{ (m_1, \ldots, m_n) \in \mathbb{Z}^n : \sum_{i=1}^n m_i \equiv 0 \pmod 2 \right\}. \tag{70}$$

Write

$$x_i := X_{\varepsilon_i}, \qquad L := t_l^{1/2}, \qquad S := t_s^{1/2}, \tag{71}$$

with

$$t_0 = t_1 = \cdots = t_{n-1} = t_l, \qquad t_n = t_s. \tag{72}$$

For $1 \le i < n$, s_i swaps x_i and x_{i+1}, while s_n sends x_n to x_n^{-1}. The affine reflection satisfies

$$(s_0 f)(x_1, x_2, x_3, \dots, x_n) = f\left(\frac{1}{qx_2}, \frac{1}{qx_1}, x_3, \dots, x_n\right). \tag{73}$$

Since $X_{-\alpha_0} = qx_1x_2$, the polynomial representation is

$$T_i f = Ls_i f + (L - L^{-1})\frac{f - s_i f}{1 - x_{i+1}/x_i}, \qquad 1 \le i < n, \tag{74}$$

$$T_n f = Ss_n f + (S - S^{-1})\frac{f - s_n f}{1 - x_n^{-1}}, \tag{75}$$

$$T_0 f = Ls_0 f + (L - L^{-1})\frac{f - s_0 f}{1 - qx_1x_2}. \tag{76}$$

Call $\mu \in M$ *anti-dominant* if

$$(\mu, A_i^\vee) \le 0, \qquad 1 \le i \le n. \tag{77}$$

For anti-dominant μ, Ion defines

$$Y_\mu = T_{\lambda_\mu}. \tag{78}$$

For example,

$$Y_{-\theta} = T_{s_\theta} T_0, \tag{79}$$

because $\lambda_\theta = s_0 s_\theta$ and $\lambda_{-\theta} = s_\theta s_0$.

Lemma 7.1. *For B_n, the $\mathbb{Z}$-span of the $\mathring{W}$-orbit of $\theta = \varepsilon_1 + \varepsilon_2$ is M.*

Proof. The orbit consists of the long roots $\pm\varepsilon_i \pm \varepsilon_j$, and their integer span is precisely the even-sum lattice M. □

Consequently, for any $\mu \in M$, one can choose

$$\mu = \sum_{k=1}^{N} m_k w_k(\theta), \qquad m_k \in \mathbb{Z}, \quad w_k \in \mathring{W}. \tag{80}$$

Since translations commute,

$$\boxed{\lambda_\mu = \prod_{k=1}^{N} \left(w_k s_0 s_\theta w_k^{-1}\right)^{m_k}.} \tag{81}$$

Every w_k, w_k^{-1}, and s_θ can be expressed using $s_1, \dots, s_n$. Thus this is an expression for λ_μ in the affine Weyl group using $s_0, s_1, \dots, s_n$.

Remark 7.2 (Reduced expressions are essential). *The preceding identity is an identity in the affine Weyl group. A non-reduced expression cannot in general be lifted by simply replacing every s_i with T_i, because $s_i^2 = 1$ whereas*

$$T_i^2 = (t_i^{1/2} - t_i^{-1/2})T_i + 1.$$

For anti-dominant μ, one must first choose a reduced expression for λ_μ and then take its positive braid lift. For general μ, one may write $\mu = \mu_1 - \mu_2$ with μ_1, μ_2 anti-dominant and use $Y_\mu = Y_{\mu_1} Y_{\mu_2}^{-1}$.

7.3 Complete computation of $E_{\varepsilon_1}^{B_n^{(1)}}$

Lemma 7.3 (Triangular support of ε_1). *With respect to (33),*

$$\{\eta \in \mathring{Q} : \eta \preceq \varepsilon_1\} = \{\varepsilon_1, 0\}. \tag{82}$$

Consequently,

$$\mathcal{P}_{\preceq \varepsilon_1} = \mathrm{span}_{\mathbb{F}}\{x_1, 1\}. \tag{83}$$

Proof. If $\eta_+ \lneq \varepsilon_1$ in dominance order, the minimality of ε_1 among nonzero dominant elements of $\mathring{Q} = \mathbb{Z}^n$ forces $\eta_+ = 0$, hence $\eta = 0$. It remains to inspect $\mathring{W}\varepsilon_1 = \{\pm\varepsilon_i : 1 \leq i \leq n\}$. For the shortest element $v(\eta)$ sending η to the anti-dominant weight $-\varepsilon_1$,

$$\ell(v(\varepsilon_i)) = 2n - i, \qquad \ell(v(-\varepsilon_i)) = i - 1. \tag{84}$$

Thus $\ell(v(\varepsilon_1)) = 2n - 1$ is the unique maximum. Since $u \leq w$ in Bruhat order implies $\ell(u) \leq \ell(w)$, the condition $v(\varepsilon_1) \leq v(\eta)$ is possible only when $\eta = \varepsilon_1$. □

It follows that

$$E_{\varepsilon_1}^{B_n^{(1)}} = E_{\varepsilon_1}^{\mathrm{I}} = x_1 + c \tag{85}$$

contains a single unknown coefficient.

Take the anti-dominant translation indexed by $-\theta = -\varepsilon_1 - \varepsilon_2$. Put

$$r_2 = s_2 s_3 \cdots s_{n-1} s_n s_{n-1} \cdots s_3 s_2. \tag{86}$$

Since $r_2(\alpha_1) = \theta$,

$$s_\theta = r_2 s_1 r_2. \tag{87}$$

Define

$$R_2 = T_2 T_3 \cdots T_{n-1} T_n T_{n-1} \cdots T_3 T_2. \tag{88}$$

Then Ion's formula gives the reduced positive braid lift

$$Y_{-\theta} = R_2 T_1 R_2 T_0. \tag{89}$$

From (73),

$$s_0 x_1 = q^{-1} x_2^{-1}, \qquad \frac{x_1 - q^{-1} x_2^{-1}}{1 - q x_1 x_2} = -q^{-1} x_2^{-1}. \tag{90}$$

Therefore

$$\begin{aligned} T_0 x_1 &= Lq^{-1}x_2^{-1} + (L - L^{-1})(-q^{-1}x_2^{-1}) \\ &= q^{-1}L^{-1}x_2^{-1}. \end{aligned} \tag{91}$$

Acting from right to left in (89) gives

$$R_2 T_0 x_1 = q^{-1}L^{-(2n-3)}S^{-1}x_2 + q^{-1}L^{-1}(S^{-1} - S), \tag{92}$$

$$T_1 R_2 T_0 x_1 = q^{-1}L^{-(2n-2)}S^{-1}x_1 + q^{-1}(S^{-1} - S). \tag{93}$$

In R_2, the long-root generators occur $2n-4$ times and the short-root generator occurs once. All the corresponding reflections fix x_1, so

$$R_2 x_1 = L^{2n-4}Sx_1, \qquad R_2(1) = L^{2n-4}S. \tag{94}$$

Substitution into (93) gives

$$\begin{aligned} Y_{-\theta} x_1 &= q^{-1}L^{-(2n-2)}S^{-1}(L^{2n-4}S)x_1 + q^{-1}(S^{-1} - S)L^{2n-4}S \\ &= q^{-1}L^{-2}x_1 + q^{-1}L^{2n-4}(1 - S^2). \end{aligned} \tag{95}$$

Hence

$$\boxed{Y_{-\theta} x_1 = q^{-1}t_l^{-1}x_1 - q^{-1}t_l^{n-2}(t_s - 1).} \tag{96}$$

Each T_i satisfies $T_i 1 = t_i^{1/2}1$. In (89), the long-root generators occur $4n-6$ times in total and the short-root generators occur twice. Thus

$$\boxed{Y_{-\theta} 1 = t_l^{2n-3}t_s.} \tag{97}$$

Define

$$a_B := q^{-1}t_l^{-1}, \qquad b_B := -q^{-1}t_l^{n-2}(t_s - 1), \qquad d_B := t_l^{2n-3}t_s. \tag{98}$$

Then

$$\gamma_{\varepsilon_1}^{B_n^{(1)}}(-\theta) = a_B, \qquad \gamma_0^{B_n^{(1)}}(-\theta) = d_B. \tag{99}$$

Assume $a_B \neq d_B$. Inserting (85) into the eigenvalue equation and comparing constants gives

$$b_B + c d_B = c a_B. \tag{100}$$

Therefore

$$\boxed{c_B := c = \frac{q^{-1}t_l^{n-2}(t_s - 1)}{t_l^{2n-3}t_s - q^{-1}t_l^{-1}}.} \tag{101}$$

We have obtained

$$\boxed{E_{\varepsilon_1}^{B_n^{(1)}} = E_{\varepsilon_1}^{\mathrm{I}} = x_1 + \frac{q^{-1}t_l^{n-2}(t_s - 1)}{t_l^{2n-3}t_s - q^{-1}t_l^{-1}}.} \tag{102}$$

Proposition 7.4 (Common eigenvector property). *For generic parameters, the polynomial in* (102) *is a common eigenvector of the full commuting Y-family.*

Proof. By Lemma 7.3 and Theorem 5.1,

$$V_{\varepsilon_1} := \mathcal{P}_{\leq \varepsilon_1} = \operatorname{span}_{\mathbb{F}}\{x_1, 1\} \tag{103}$$

is invariant under every Y_μ. In the ordered basis $\{x_1, 1\}$,

$$Y_{-\theta}\big|_{V_{\varepsilon_1}} = \begin{pmatrix} a_B & 0 \\ b_B & d_B \end{pmatrix}. \tag{104}$$

When $a_B \neq d_B$, its a_B-eigenspace is one-dimensional and is spanned by

$$E = x_1 + \frac{b_B}{a_B - d_B} = x_1 + c_B. \tag{105}$$

For $\mu \in M$, the restriction $Y_\mu|_{V_{\varepsilon_1}}$ commutes with $Y_{-\theta}|_{V_{\varepsilon_1}}$. Hence

$$Y_{-\theta}(Y_\mu E) = Y_\mu(Y_{-\theta} E) = a_B Y_\mu E. \tag{106}$$

Thus $Y_\mu E$ lies in the one-dimensional a_B-eigenspace, so there is a scalar $\gamma_{\varepsilon_1}^{B_n^{(1)}}(\mu)$ such that

$$Y_\mu E = \gamma_{\varepsilon_1}^{B_n^{(1)}}(\mu) E. \tag{107}$$

The leading term is x_1, so triangular uniqueness identifies E with $E_{\varepsilon_1}^{B_n^{(1)}}$. □

7.4 The corresponding polynomial of type $C_n^{(1)}$

Set

$$\eta_i := \frac{\varepsilon_i}{\sqrt{2}}, \qquad \beta_0 := \frac{\theta}{\sqrt{2}} = \eta_1 + \eta_2 \in \mathring{Q}^\iota, \qquad y_i := X^\iota_{\eta_i}. \tag{108}$$

Although an individual y_i does not belong to $\mathbb{F}[\mathring{Q}^\iota]$, every monomial index that actually occurs below lies in $\mathring{Q}^\iota$. The embedding into $\mathbb{F}[y_1^{\pm 1}, \ldots, y_n^{\pm 1}]$ is only a computational notation.

The highest long root of the scaled C_n system is

$$\theta^\iota = 2\eta_1 = \sqrt{2}\,\varepsilon_1, \tag{109}$$

and $-2\eta_1 \in M^\iota$ is anti-dominant. The finite reflection has the reduced expression

$$s_{2\eta_1} = s_1 s_2 \cdots s_{n-1} s_n s_{n-1} \cdots s_2 s_1. \tag{110}$$

Consequently, Ion's Remark 1.4 gives

$$\boxed{Y^{C_n^{(1)}}_{-2\eta_1} = T_1 T_2 \cdots T_{n-1} T_n T_{n-1} \cdots T_2 T_1 T_0.} \tag{111}$$

Under the parameter-preserving map ω_R, the short-root parameter of the dual C_n system is t_l, while its long-root parameter is t_s. Thus

$$L = t_l^{1/2} = (t^C_{\mathrm{sh}})^{1/2}, \qquad S = t_s^{1/2} = (t^C_{\mathrm{lg}})^{1/2}. \tag{112}$$

The affine simple root is

$$\alpha_0^\iota = \delta - 2\eta_1, \tag{113}$$

so

$$s_0(y_1) = q^{-1}y_1^{-1}, \qquad X^\iota_{-\alpha_0^\iota} = qy_1^2. \tag{114}$$

Using

$$T_i X_\lambda = t_i^{1/2} X_{s_i\lambda} + (t_i^{1/2} - t_i^{-1/2})\frac{X_\lambda - X_{s_i\lambda}}{1 - X_{-\alpha_i}}, \tag{115}$$

and acting from the rightmost end of (111), set

$$U := T_2T_3\cdots T_{n-1}T_nT_{n-1}\cdots T_3T_2. \tag{116}$$

The intermediate steps are

$$T_0(y_1y_2) = q^{-1}S^{-1}y_1^{-1}y_2, \tag{117}$$

$$T_1T_0(y_1y_2) = q^{-1}S^{-1}\left(L^{-1}y_1y_2^{-1} + L^{-1} - L\right), \tag{118}$$

$$UT_1T_0(y_1y_2) = q^{-1}\left(L^{-(2n-3)}S^{-2}y_1y_2 + L^{2n-5} - L^{2n-3}\right), \tag{119}$$

$$T_1UT_1T_0(y_1y_2) = q^{-1}\left(L^{-(2n-4)}S^{-2}y_1y_2 + L^{2n-4} - L^{2n-2}\right). \tag{120}$$

The last line is $Y^{C_n^{(1)}}_{-2\eta_1}X^\iota_{\beta_0}$. Therefore

$$\boxed{Y^{C_n^{(1)}}_{-2\eta_1}X^\iota_{\beta_0} = q^{-1}t_l^{-(n-2)}t_s^{-1}X^\iota_{\beta_0} + q^{-1}t_l^{n-2}(1-t_l).} \tag{121}$$

In (111), the short-root generators occur $2n-2$ times, while the long-root generators T_0, T_n occur twice. Hence

$$\boxed{Y^{C_n^{(1)}}_{-2\eta_1}1 = t_l^{n-1}t_s.} \tag{122}$$

The weight β_0 is the minimal nonzero dominant element of $\mathring{Q}^\iota$. The same triangular-support argument therefore gives

$$E^{C_n^{(1)}}_{\beta_0} = X^\iota_{\beta_0} + c_C. \tag{123}$$

Set

$$a_C := q^{-1}t_l^{-(n-2)}t_s^{-1}, \qquad d_C := t_l^{n-1}t_s. \tag{124}$$

Equations (121)–(122) say

$$\gamma^{C_n^{(1)}}_{\beta_0}(-2\eta_1) = a_C, \qquad \gamma^{C_n^{(1)}}_{0}(-2\eta_1) = d_C. \tag{125}$$

Solving the one-dimensional eigenvector equation yields

$$\boxed{E^{C_n^{(1)}}_{\beta_0} = X^\iota_{\beta_0} + c_C, \qquad c_C = \frac{q^{-1}t_l^{n-2}(t_l-1)}{t_l^{n-1}t_s - q^{-1}t_l^{-(n-2)}t_s^{-1}}.} \tag{126}$$

7.5 An explicit B_n–C_n evaluation-duality identity

Take

$$\lambda = \varepsilon_1, \qquad \beta_0 = \frac{\varepsilon_1 + \varepsilon_2}{\sqrt{2}}. \tag{127}$$

Since $\beta_0 = \theta/\sqrt{2}$, the definitions of the spectral points give

$$X^\iota_{\beta_0}\big(\mathrm{sp}_R(\varepsilon_1)\big) = \gamma^R_{\varepsilon_1}(-\theta) = a_B, \tag{128}$$

$$X^\iota_{\beta_0}\big(\mathrm{sp}_R(0)\big) = \gamma^R_0(-\theta) = d_B. \tag{129}$$

On the other side, $-\sqrt{2}\varepsilon_1 = -2\eta_1 = -\theta^\iota$, so

$$x_1\big(\mathrm{sp}_{R^\iota}(\beta_0)\big) = \gamma^{R^\iota}_{\beta_0}(-\theta^\iota) = a_C, \tag{130}$$

$$x_1\big(\mathrm{sp}_{R^\iota}(0)\big) = \gamma^{R^\iota}_0(-\theta^\iota) = d_C. \tag{131}$$

Here $x_1 = X_{\varepsilon_1}$ is the B_n-side coordinate; it should not be confused with the formal dual-side symbols y_i.

Consequently,

$$\mathcal{E}^R_{\varepsilon_1}\big(\mathrm{sp}_{R^\iota}(\beta_0)\big) = \frac{a_C + c_B}{d_C + c_B}, \tag{132}$$

whereas

$$\mathcal{E}^{R^\iota}_{\beta_0}\big(\mathrm{sp}_R(\varepsilon_1)\big) = \frac{a_B + c_C}{d_B + c_C}. \tag{133}$$

Thus general non-symmetric evaluation-duality becomes the explicit identity

$$\boxed{\frac{a_C + c_B}{d_C + c_B} = \frac{a_B + c_C}{d_B + c_C}.} \tag{134}$$

Substituting (98), (101), (124), and (126) gives

$$\frac{q^{-1}t_l^{-(n-2)}t_s^{-1} + c_B}{t_l^{n-1}t_s + c_B} = \frac{q^{-1}t_l^{-1} + c_C}{t_l^{2n-3}t_s + c_C}. \tag{135}$$

Both sides simplify to

$$\boxed{\frac{q^{-1}t_l^{n-2}(t_l + t_s - 1) - q^{-2}t_l^{-(n-1)}t_s^{-1}}{t_l^{3n-4}t_s^2 - q^{-1}t_l^{n-2}}.} \tag{136}$$

This explicitly verifies normalized evaluation-duality for the nontrivial pair

$$\lambda = \varepsilon_1, \qquad \beta_0 = \frac{\varepsilon_1 + \varepsilon_2}{\sqrt{2}}. \tag{137}$$

8 Guiding Principle

The two commutative parts of DAHA have different interpretations:

$$X_\beta \quad \text{acts as a coordinate or multiplication operator,} \tag{138}$$

whereas

$$Y_\mu \quad \text{acts as a spectral or } q\text{-difference operator.} \tag{139}$$

The DAHA involution exchanges these roles:

$$\boxed{\text{coordinate variables} \longleftrightarrow \text{spectral variables.}} \tag{140}$$

This exchange is the algebraic origin of Macdonald duality.

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